NextAIRE

NextAIRE

Next Generation AI Researchers for Air Quality Excellence

Funded by
the European Union

NextAIRE

AQE – Air Quality and Exposure

Module 1: **Foundational Air Quality Concepts**

Sabina Žero, UNSA, Sarajevo, BiH

Funded by
the European Union

Foundational Air Quality Concepts

- The module is focused to provide air quality basic concepts; terms, definisions and facts for understanding how natural processes and human activity alters the atmosphere.

- Pollutant types (particles, gases)
- Focuse on:
- physical/chemical properties,
- primary/secondary distinction,
- health and environmental impacts.

Funded by
the European Union

NextAIRE

The Air We Breathe

- Nearly 2.4 billion people still rely on polluting fuels to cook and heat their homes, leading to 3.2 million premature deaths each year from household air pollution.
- At the same time, over 99 % of the global population breathes air that exceeds WHO quality guidelines, with ambient pollution responsible for an additional 4.2 million deaths annually.

Funded by
the European Union

NextAIRE

The Air We Breathe

- Over the past twenty years, declining emissions have significantly enhanced European air quality.
- Notably, PM2,5-related deaths dropped by 57 % between 2005 and 2023, meeting the EU’s "zero pollution" target seven years early.

Funded by
the European Union

NextAIRE

The Air We Breathe

330,000

- Premature deaths in Europe during 2023 alone due to pollution levels exceeding WHO quality guidelines.

Funded by
the European Union

NextAIRE

While Europe is breathing easier, the air is not "clean" just yet.

Here is the breakdown of the current situation

Funded by
the European Union

NextAIRE

The Good News: PM2,5-linked fatalities plummeted by 57 % since 2005, hitting the EU's 2030 goal way ahead of schedule.

The Reality: Air pollution remains the No. 1 environmental health risk in the region.

The Human Cost (2023): Exposure above WHO limits caused hundreds of thousands of premature deaths:

206,000

71,000

56,000

Funded by
the European Union

Atmosphere constituents

Composition of Earth's atmosphere

Major components (dry air) and trace gases. Data representative of the lower troposphere.

Major components (dry air)

Nitrogen (N_2) ~78.08%
Dilutes O_2; essential for protein synthesis

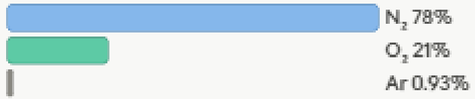
Oxygen (O_2) ~20.95%
Sustains aerobic respiration and combustion

Argon (Ar) ~0.93%
Chemically inert noble gas; radiogenic origin

Dry air total: $N_2 + O_2 + Ar = 99.96\%$

Water vapour (H_2O) 0 – 4%
Variable component; controls latent heat transport and cloud formation

Proportional composition (dry air)



Trace gases (ppm by volume)

Species	Concentration	Class
Carbon dioxide (CO_2)	~425 ppm	GHG
Methane (CH_4)	~1.93 ppm	GHG
Nitrous oxide (N_2O)	~0.33 ppm	GHG / ODS
Ozone (O_3)	~0.07 ppm *	GHG / UVB
Neon (Ne)	~18 ppm	Noble
Helium (He)	~5 ppm	Noble
Hydrogen (H_2)	~0.5 ppm	Reactive
Krypton (Kr)	~1 ppm	Noble
Xenon (Xe)	~0.09 ppm	Noble
Other (CO, NH_3, SO_2, …)	< 0.05 ppm	Various

GHG = greenhouse gas ODS = ozone-depleting substance
UVB = absorbs ultraviolet-B radiation
* Tropospheric mean; stratospheric values reach ~10 ppm

Note: Values represent the well-mixed troposphere (0–12 km). Composition above the tropopause differs significantly; the stratosphere (12–50 km) contains the ozone layer (peak ~3.5 ppm at ~25 km altitude).

Figure 1. Approximate composition of Earth's lower atmosphere. CO_2 value reflects early 2026 measurements (NOAA Global Monitoring Laboratory). All other values are long-term dry-air averages.

Funded by
the European Union

Atmosphere constituents

These abundances are regulated over geologic timescales by:

- Biosphere activity
- Uptake/release from crustal material
- Degassing of the Earth's interior

Funded by
the European Union

Atmosphere constituents

Water vapor is the next most abundant constituent:

- Found mostly in the lower atmosphere
- Highly variable, up to 3 % concentration
- Controlled by evaporation and precipitation

Trace gases (<1 % of atmosphere):

- Crucial for Earth's radiative balance and atmospheric chemistry
- Have changed rapidly and significantly over the last two centuries

Funded by
the European Union

Important facts

The atmosphere receives many products of technological society.

Effluents include:

- Combustion products from fossil fuels
- Newly developed synthetic chemicals

Historically, emissions have led to unforeseen consequences:

- 1950s: Motor vehicle emissions → urban smog
- 1970s: Chlorofluorocarbon (CFC) emissions → stratospheric ozone depletion

Concentrations commonly expressed in mass per unit volume ($\mu g/m^3$) or a volume mixing ratio (1 ppm= 10^{-6} v/v, 1 ppb= 10^{-9} v/v)

Funded by
the European Union

NextAIRE

Important facts

Shift in focus (late 19th–early 20th century):

- From major atmospheric constituents → trace constituents (<1 ppm by volume)

Trace species in the atmosphere:

- Present due to geologic, biological, chemical, and anthropogenic processes
- Atmosphere contains a myriad of such species

Advances in instrumentation (since ~1975):

- Enabled detection down to 1 part per trillion (ppt) by volume

Innovations improved identification of trace species

Funded by
the European Union

The amount of substances in the atmosphere -Units

Global observations:

- Atmospheric composition is changing on a worldwide scale

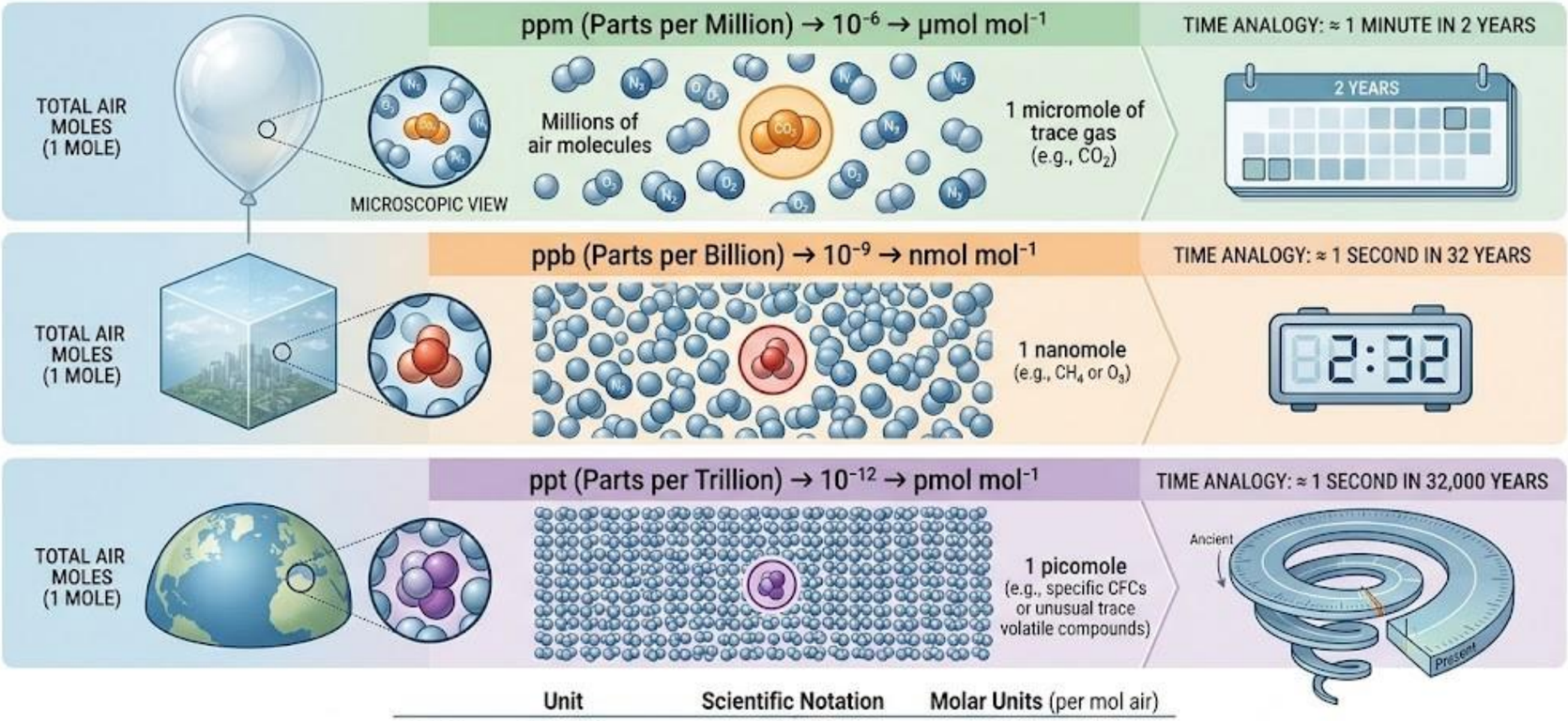
Mixing ratios in atmospheric chemistry are commonly expressed as:

- Parts per million (ppm) → 10^{-6} → $\mu mol\ mol^{-1}$
- Parts per billion (ppb) → 10^{-9} → $nmol\ mol^{-1}$
- Parts per trillion (ppt) → 10^{-12} → $pmol\ mol^{-1}$

Funded by
the European Union

NextAIRE

ATMOSPHERIC MIXING RATIOS: VISUALIZING PARTS PER NOTATION



	Unit	Scientific Notation	Molar Units (per mol air)
ppm	1×10^{-6}	1×10^{-6}	μmol mol^{-1}
ppb	1×10^{-12}	1×10^{-9}	pmol mol^{-1}

Funded by
the European Union

Photochemical Processes in the Atmosphere

Solar radiation effects in atmospheric chemistry

- Leads to photolysis of trace gases
- Causes photooxidation of oxidizable trace gases in the troposphere

Most significant feature

- Photochemical reactions driven by absorption of photons
- Radiation comes from the sun, mainly in the ultraviolet (UV) part of the electromagnetic (EM) spectrum

Funded by
the European Union

Photochemical Processes in the Atmosphere

Photochemical Initiation

- The Photon Factor: Photochemical reactions are triggered when a molecule absorbs a photon ($h\nu$)
- Ozone Decomposition: Ozone (O_3) absorbs ultraviolet radiation to break apart
- Energy Dynamics: Ultraviolet (UV) radiation has a higher frequency than visible light, making it energetic enough to rupture chemical bonds

Excited States

- Definition: Products of photochemical reactions are often "electronically energized," denoted by an asterisk (*).
- Chemical Reactivity: This excess electronic energy makes the species significantly more reactive than its ground-state counterpart

Funded by
the European Union

Photochemical Processes in the Atmosphere

Free Radicals

- Definition: Highly reactive species containing an unpaired electron, typically denoted with a dot
- Hydroxyl Radical Production: The reaction of an excited oxygen atom with water vapor
- Atmospheric Impact: Due to the strong tendency for electrons to pair up, free radicals drive the majority of critical chemical processes in the atmosphere.

Funded by
the European Union

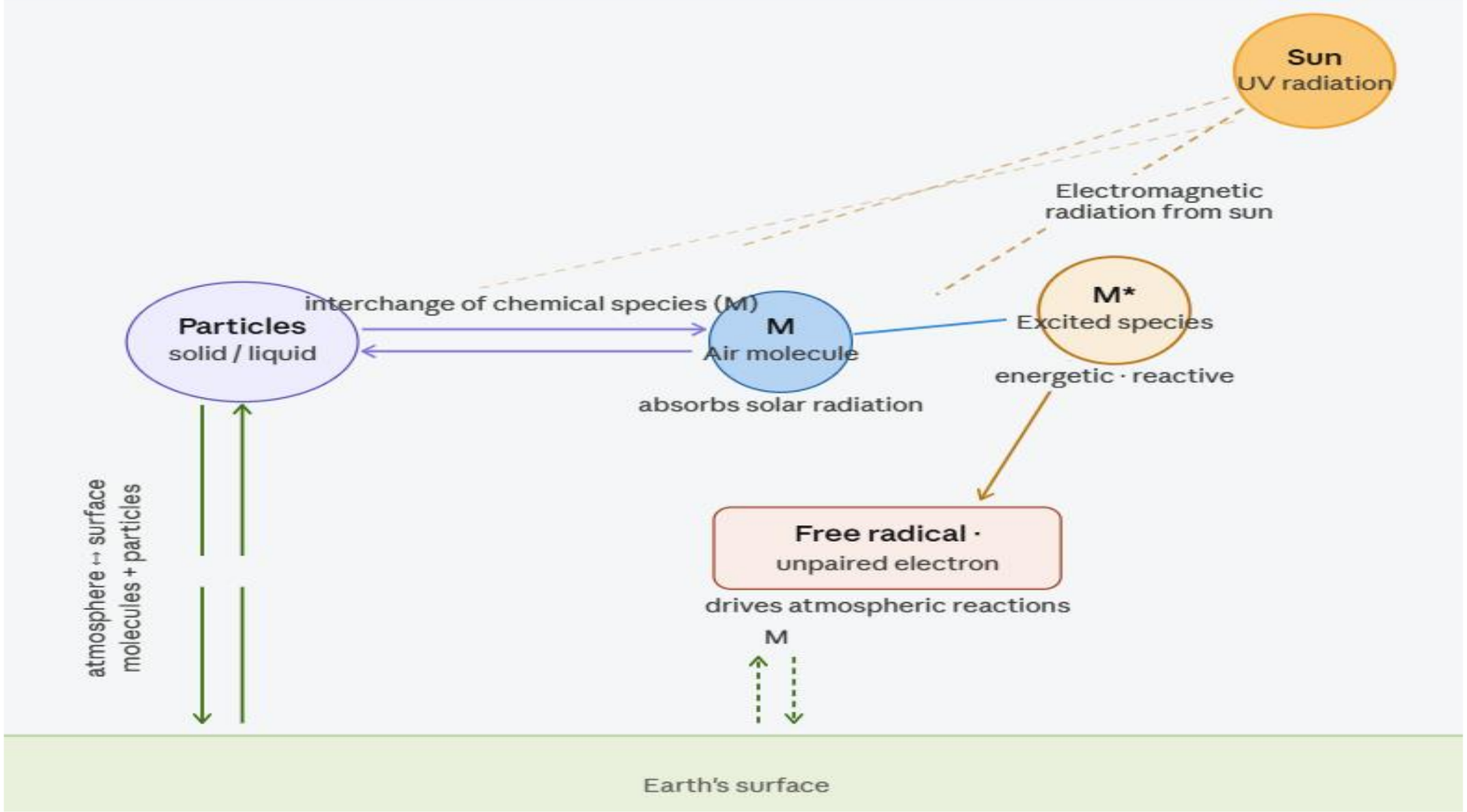
Photochemical Processes in the Atmosphere

The "Third Body" (M)

- Definition: A chemical species that participates in a collision without being chemically altered.
- Ozone Formation: Essential for producing stratospheric ozone.
- Energy Stabilization: The third body absorbs excess energy released during the bond formation. Without M, the energy would cause the newly formed O_3 to immediately fall apart.

Funded by
the European Union

NextAIRE



The major atmospheric chemical processes
(adapted from Manahan S.E., Environmental Chemistry)

Funded by
the European Union

NextAIRE

Air pollution

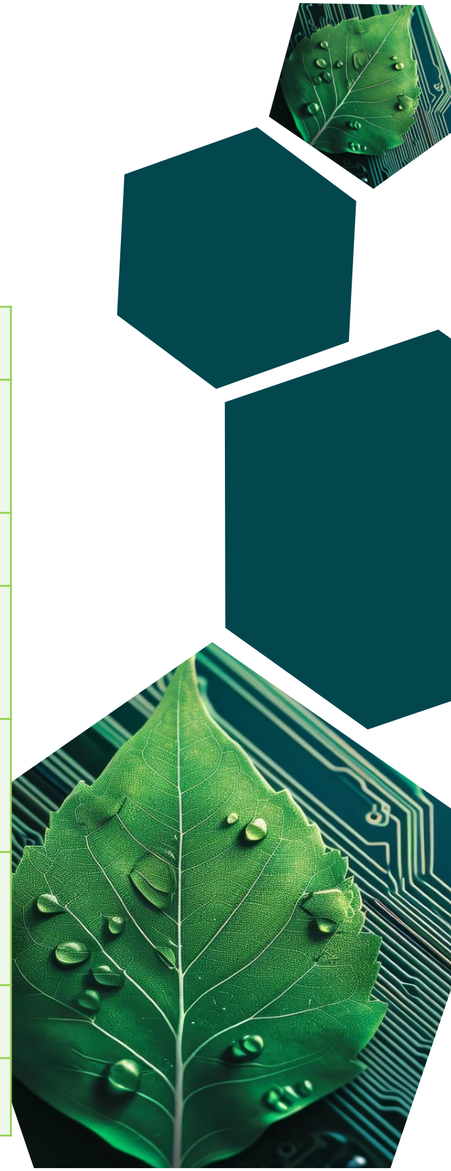
- Air pollution in the outdoor environment, that is, in outdoor air, but which can enter or be present in indoor environments.
- Outdoor air pollution is a major environmental health problem affecting everyone in low-, middle-, and high-income countries.

Funded by
the European Union

NextAIRE

Air pollution

Source	Ambient Air Pollution	Household Air Pollution
Primary driver	High-temperature combustion	Incomplete combustion from domestic activities
Transport	Major contributor	Minimal
Industry & energy	Major contributor	Minimal
Cooking	Minimal	Major contributor (inefficient technologies)
Heating	Minimal	Major contributor (inefficient technologies)
Lighting	Minimal	Kerosene use
Routine tasks	Minimal	Boiling water, preparing animal feed



Funded by
the European Union

NextAIRE

Air Pollutants

Air pollutants - substances introduced into the atmosphere either by human activity or natural processes, that have harmful effects on human health, ecosystems, or the built environment.

Air pollutants exist as gases, liquids, or solid particles.

Natural vs. anthropogenic sources: It distinguishes between organic compounds emitted by plants (biogenic VOCs like isoprene and terpenes) and those from human activity (fuel evaporation, incomplete combustion, and industrial solvents).

Funded by
the European Union

Air Pollution

Pollutants with the strongest evidence for public health concern include:

- particulate matter (PM),
- carbon monoxide (CO),
- ozone (O_3),
- nitrogen dioxide (NO_2) and
- sulfur dioxide (SO_2)
- organic compounds (PAH, VOCs)

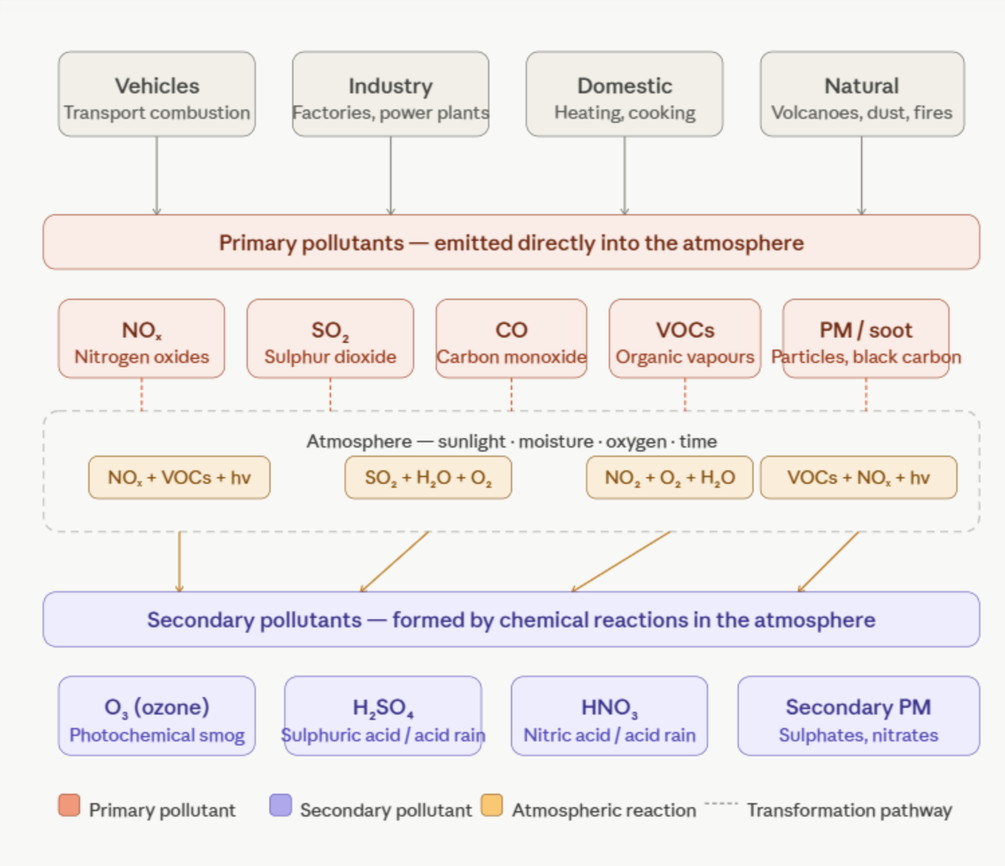
Health problems can occur as a result of both short- and long-term exposure to these various pollutants. For some pollutants, there are no thresholds below which adverse effects do not occur.

Funded by
the European Union

Primary/Secondary Pollutants

Primary Pollutants: Emitted directly from a source

Secondary Pollutants: Formed in the atmosphere through chemical reactions between primary pollutants and natural components



Funded by
the European Union

NextAIRE

Air Pollutants Major Sources

Particulate matter (PM)
Combustion, dust, industrial emissions

Nitrogen oxides (NO_x)
Vehicle engines, power plants

Sulphur oxides (SO_x)
Coal burning, volcanic activity

Carbon oxides (CO, CO_2)
Incomplete combustion, transport

Volatile organics (VOCs)
Fuels, solvents, paints

Heavy metals
Industrial processes, mining

Funded by the European Union

NextAIRE

Air Pollutants

Category	Examples	Key Sources
Particulate matter (PM)	PM2,5, PM10, black carbon	Combustion, dust, industrial emissions
Nitrogen oxides (NOx)	NO, NO_2	Vehicle engines, power plants
Sulphur oxides (SOx)	SO_2, SO_3	Coal burning, volcanic activity
Carbon oxides	CO, CO_2	Incomplete combustion, transport
Ozone (O_3)	Ground-level ozone	Secondary pollutant formed from NOx + VOCs + sunlight
Volatile organic compounds (VOCs)	Benzene, formaldehyde	Fuels, solvents, paints
Heavy metals	Lead, mercury, arsenic	Industrial processes, mining
Persistent organic pollutants (POPs)	Dioxins, PCBs	Waste incineration, pesticides

Funded by
the European Union

Particulate matter, PM10

- Particulate matter with a diameter of 10 µm or less (PM10) originates from a mix of anthropogenic and natural sources.
- While domestic heating remains the primary driver, industrial and agricultural activities also contribute significantly.
- Additionally, secondary PM10 forms through atmospheric chemical reactions.
- The major components of PM are sulfates, nitrates, ammonia, sodium chloride, black carbon, mineral dust and water.

Funded by
the European Union

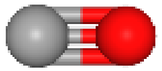
Particulate matter, PM2,5

- Fine particulate matter (PM2,5) consists of particles with a diameter of 2,5 µm or less.
- The primary drivers of these emissions are domestic solid-fuel heating, industrial operations, and road transportation.
- Similar to PM10, PM2,5 is produced through both natural processes and atmospheric chemical reactions; notably agricultural ammonia emissions play a critical role in the secondary formation of these particles.

Funded by
the European Union

Carbon monoxide, CO

- Formation: Produced by the incomplete combustion of carbon-containing fuels
- The "Silent Killer": It is a colorless, odorless gas that is toxic because it binds to hemoglobin in the blood much more strongly than oxygen, preventing oxygen transport



Funded by
the European Union

Ozone, O_3

- Ozone at ground level – not to be confused with the ozone layer in the upper atmosphere – is one of the major constituents of photochemical smog
- It is formed through the reaction with gases in the presence of sunlight
- Synthesized in the atmosphere through photochemical reactions between nitrogen oxides and volatile organic compounds (VOCs), including methane
- These precursors originate from both anthropogenic and biogenic sources.

Funded by
the European Union

Ozone, O_3

- Beyond local emissions, European ozone levels are influenced by long-range atmospheric transport from the northern hemisphere and the stratosphere
- This chemical process is driven by heat and sunlight, meteorology is the primary factor in the significant inter annual fluctuations observed in ozone concentrations
- Exposure to excessive ozone can cause problems breathing, trigger asthma, reduce lung function and lead to lung disease

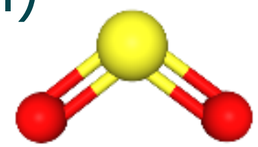
Funded by
the European Union

Oxides of nitrogen, NO_x

- The Components: Primarily focuses on nitric oxide and nitrogen dioxide
- Nitrogen dioxide (NO_2) is a reddish-brown gas that is soluble in water, and a strong oxidant.
- Atmospheric Impact: NOx is a critical "ingredient" for the formation of photochemical smog and secondary particulate matter
- Source: Formed during high-temperature combustion (like in car engines) when atmospheric nitrogen and oxygen react
- Health & Environment: NO_2 is a respiratory irritant closely linked to asthma and other respiratory conditions and contributes to the formation of nitric acid, a major component of acid rain, is an important ozone precursor

Funded by
the European Union

Sulfur dioxide, SO_2



- Primary Source: Mainly enters the atmosphere through the combustion of sulfur-containing fossil fuels (especially coal) and the smelting of metal ores
- Chemical Transformation: SO_2 can be oxidized in the atmosphere to form sulfuric acid (H_2SO_4) which can then associate with particles to form corrosive sulfate mists
- Health Risk: It is a heavy respiratory irritant that can cause permanent lung damage at high concentrations and is toxic to plant life

Funded by
the European Union

NextAIRE

Polycyclic Aromatic Hydrocarbons, PAHs

- Byproducts of the incomplete combustion of organic materials and fuels.
- Key sources include coal, crude and heavy fuel oils, and biofuels like firewood and agricultural waste, as well as tobacco smoke.
- Over the last three to four decades, the proportional contribution of these various sources to outdoor air PAH levels has shifted significantly.

Funded by
the European Union

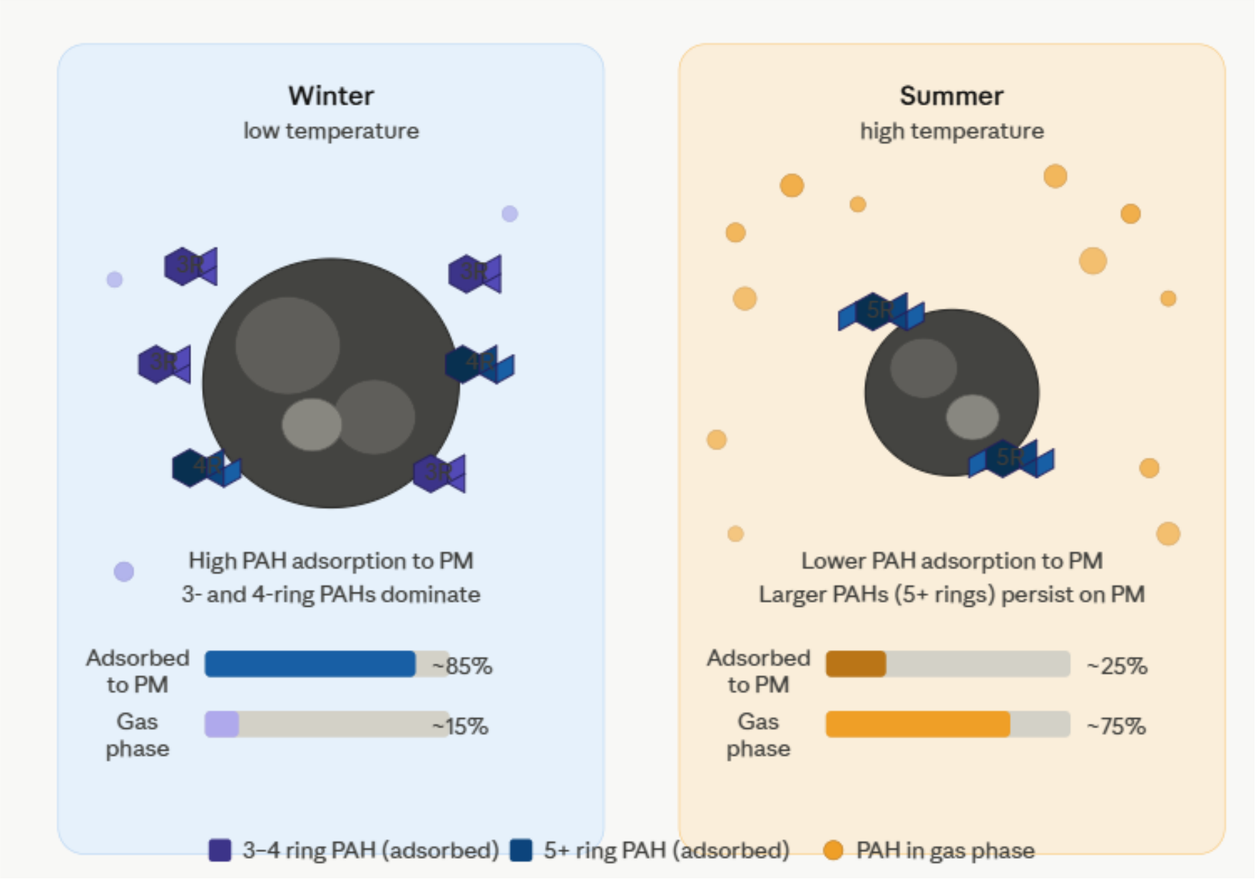
NextAIRE

Polycyclic Aromatic Hydrocarbons, PAHs

- The behavior of PAHs in the atmosphere depends largely on their structure.
- PAHs with five or more aromatic rings are lipophilic and poorly water-soluble, causing them to adsorb primarily onto the black carbon found in fine particulate matter. In contrast, PAHs with four rings or fewer usually exist in a gaseous state.
- However, because these smaller PAHs are much more abundant in outdoor air, they still represent the majority of the PAHs found on fine and ultrafine particles.

Funded by
the European Union

Polycyclic Aromatic Hydrocarbons, PAHs



Funded by
the European Union

Lead (Pb)

- Lead (Pb) and lead particulate compounds can be found in the home in contaminated dust from products such as paints, ceramics, pipes and plumbing materials, solders, gasoline, batteries, ammunition, and cosmetics. Lead can also be found in ambient air from vehicle exhaust of fuel with lead.
- Lead poses health risks of particular concern for children and pregnant women. The health impacts for children exposed to lead include behavior and learning problems, slowed growth, hearing problems, and anemia.

Funded by
the European Union

Arsenic (As), Nickel (Ni), Cadmium (Cd)

- The concentration of arsenic (As), nickel (Ni), and cadmium (Cd) in the air is a significant concern due to their potential health risks and environmental impact.
- These metals are often found in particulate matter (PM), which can be harmful to human health and the ecosystem. The concentration of these metals in the air varies depending on the source of emissions, such as industrial processes, road transport, and other sources.

Funded by
the European Union

Inorganic Particulate Matter & Ammonia

- Ammonia's Role: While a gas, NH_3 is the only major basic (alkaline) gas in the atmosphere
- Neutralization: It plays a crucial role in neutralizing acidic gases (like nitric and sulfuric acid) to form solid ammonium salts (NH_4NO_3), which are a major component of PM2,5

Funded by
the European Union

Other Inorganic Gases

- Fluorides and Chlorides: Released from industrial processes (like aluminum smelting or coal burning). Hydrogen fluoride (HF) is particularly toxic to plants.
- Hydrogen Sulfide (H_2S): A highly toxic, "rotten egg" smelling gas produced by geothermal activity and the decay of organic matter in anaerobic conditions.

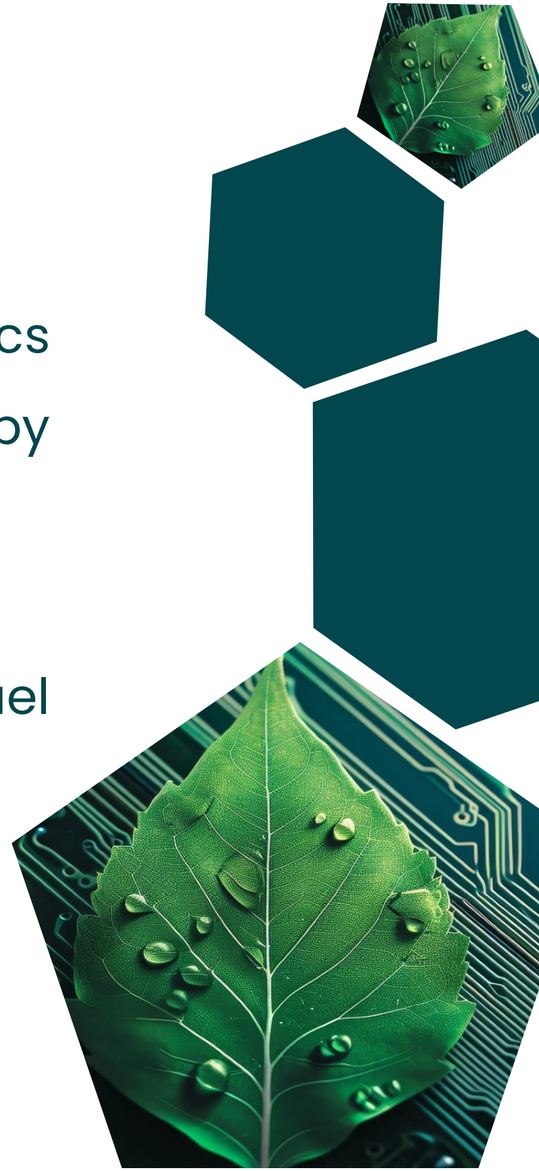
Funded by
the European Union

Organic Pollutants

- Natural (Biogenic) Sources: A significant portion of atmospheric organics comes from plants. Examples include isoprene and terpenes (emitted by trees like pines and oaks), which are highly reactive

- Anthropogenic Sources: Human-driven emissions come primarily from fuel evaporation, incomplete combustion, and the use of industrial solvents

Compound Class	Common Examples	Primary Concern
PAHs	Benzo(a)pyrene	Potent carcinogens; bind to Particulate Matter.
VOCs	Benzene, Isoprene	Precursors to ground-level ozone and smog.



Funded by
the European Union

NextAIRE

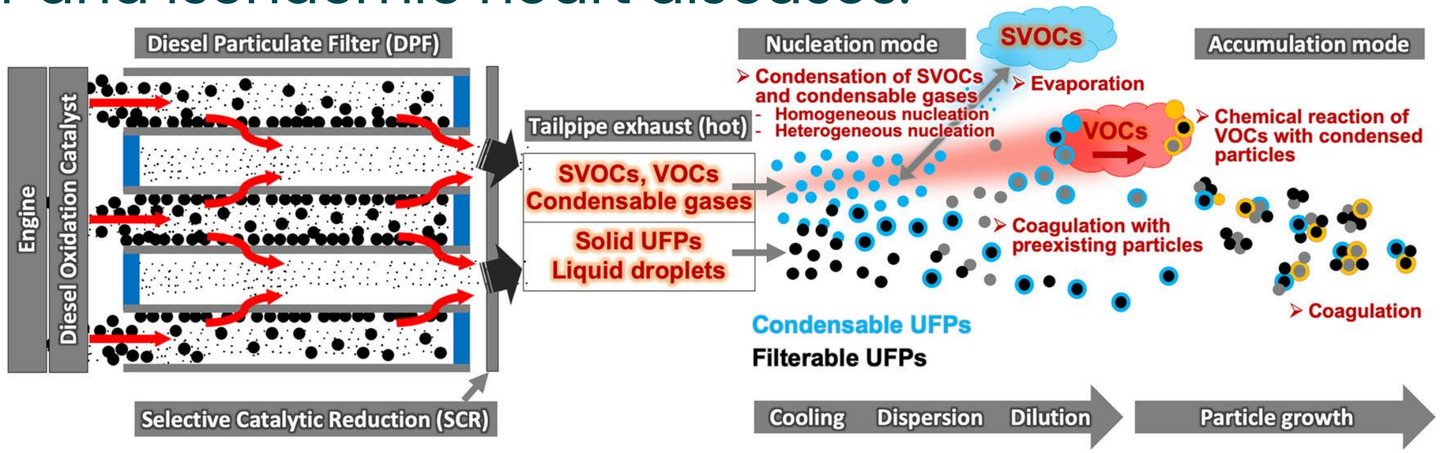
Organic Pollutants

- Benzo(a)pyrene (BaP), a potent carcinogen, originates primarily from the residential combustion of coal and wood, with additional contributions from industry and agricultural waste.
- Benzene remains in the vapor phase in the air. Benzene enters the environment through natural sources (volcanoes, forest fires) and human activities (petroleum refining, industrial manufacturing). It is a versatile industrial building block used to create everything from rubber and dyes to drugs and pesticides. Beyond its carcinogenic potential, benzene exposure is associated with several acute neurological and physical effects.

Funded by
the European Union

Ultrafine Particles

- Ultrafine particles (UFP) are particulate matter with a diameter less than or equal to 0.1 micrometer. The main source of UFP is combustion processes in transportation (e.g. vehicles, aviation, shipping), industrial and power plants and residential heating.
- Exposure to UFP can increase the likelihood of pulmonary, cardiovascular and ischaemic heart diseases.



The dynamics of ultrafine particles emitted from a typical diesel engine. Source: Kwon, HS., Ryu, M.H. & Carlsten, C. Exp Mol Med 52, 318–328 (2020).

Funded by the European Union

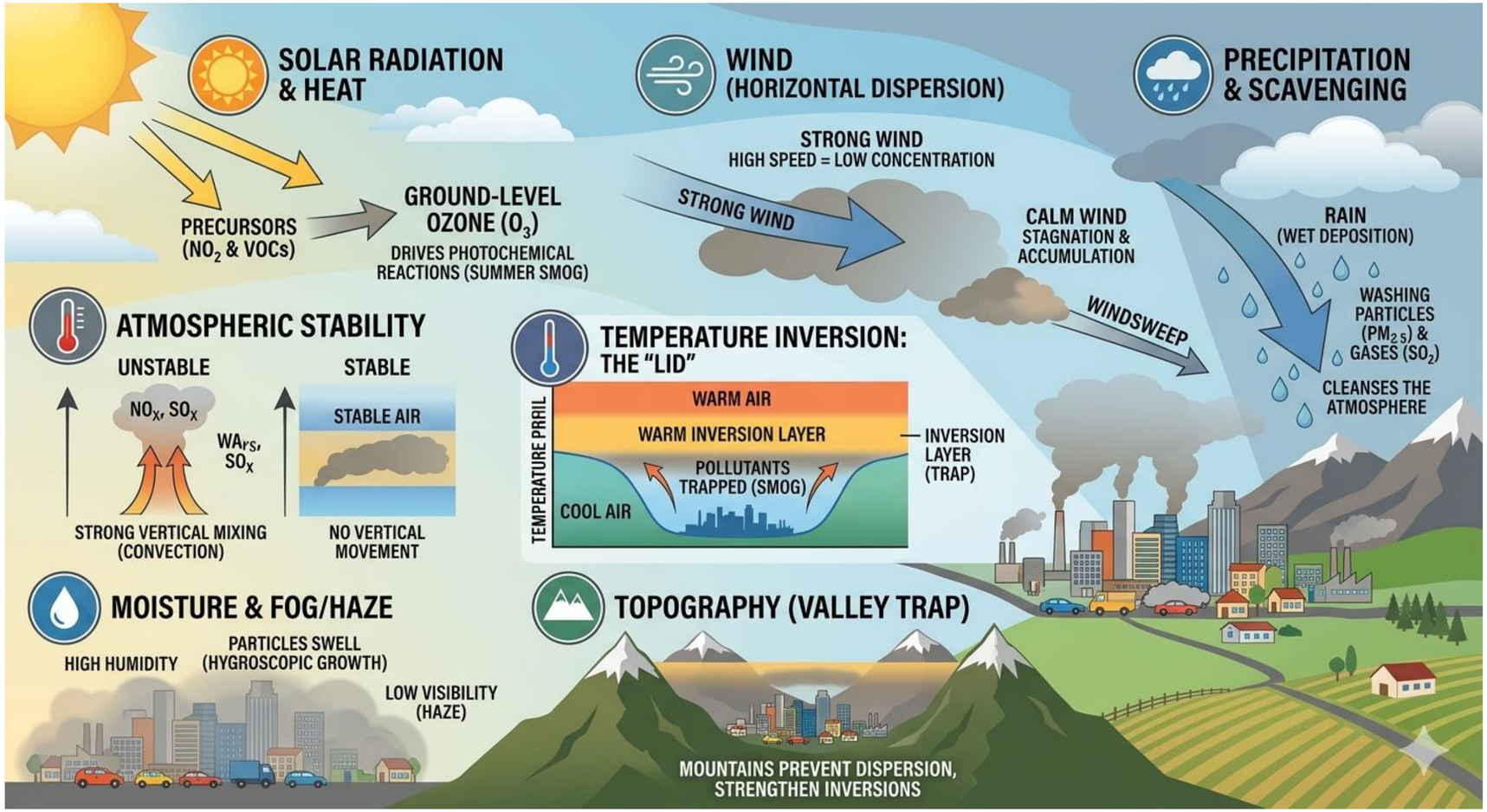
Black carbon

- Black carbon (BC) is a major component of PM2,5 and it is sometimes referred to as soot and its main sources are from incomplete combustion of fossil fuels, biofuels and biomass. It can be emitted both from anthropogenic (e.g. diesel vehicles, biomass cookstoves) and natural (e.g. wildfires) sources.
- Short- and long-term exposure to black carbon has been associated with cardiovascular health effects and premature mortality.

Funded by
the European Union

NextAIRE

Meterology and Air Pollution



Funded by
the European Union

EU air quality standards

- The EU air quality standards are set to reduce risks posed by air pollution and are outlined in the revised Ambient Air Quality Directive.
- Key updates include:
- Lower allowable levels for twelve air pollutants, including particulate matter (PM2.5 and PM10), nitrogen dioxide (NO_2), sulfur dioxide (SO_2), ozone (O_3), and others.
- The standards aim to align more closely with World Health Organization recommendations.
- The EU's clean air policy is based on three pillars: ambient air quality standards, reducing air pollution emissions, and emissions standards for key sources of pollution.

Funded by
the European Union

NextAIRE

References

- Manahan, S.E. (2022). Environmental Chemistry: Eleventh Edition (11th ed.). CRC Press. https://doi.org/10.1201/9781003096238
- Harrison, R. M. (Ed.). (2023). Pollution: causes, effects and control. Royal society of chemistry.
- Seinfeld, J. H., & Pandis, S. N. (2016). Atmospheric chemistry and physics: from air pollution to climate change. John Wiley & Sons.
- https://www.who.int/health-topics/air-pollution
- https://www.eea.europa.eu/en/topics/in-depth/air-pollution
- https://environment.ec.europa.eu/topics/air/air-quality
- Kwon, HS., Ryu, M.H. & Carlsten, C. Ultrafine particles: unique physicochemical properties relevant to health and disease. Exp Mol Med 52, 318–328 (2020). https://doi.org/10.1038/s12276-020-0405-1
- Sekar A, Varghese GK, Ravi Varma MK. Analysis of benzene air quality standards, monitoring methods and concentrations in indoor and outdoor environment. Heliyon. 2019 Nov 29;5(11):e02918. doi: 10.1016/j.heliyon.2019.e02918.
- Claude (2026). Images on slides 9,15,20,22,25,26,27,38 [AI-generated image]. Source URL: https://claude.ai
- Gemini (2026). Images on slide 47 [AI-generated image]. Source URL: https://gemini.google.com/

Funded by
the European Union

NextAIRE

What we can do to prevent further pollution and climate change?

Funded by
the European Union

NextAIRE

Consortium

Ascalia

diadikasia
BRIGHT GROUP MEMBER

smartsense

Megasense

CENTER FOR PHYSICAL SCIENCES AND TECHNOLOGY

INSTITUTE FOR ANTHROPOLOGICAL RESEARCH

theLisbonCouncil
think tank for the 21st century

INAR
Institute for Atmospheric and Earth System Research

imi

msensis

vito

KNOW Center

DHMZ

UNIVERSITAS UNIVERZITET STUDIORUM SARAJEVOENSIS USARAJEVU

DEMOKRITOS

FERIT
FACULTY OF ELECTRICAL ENGINEERING, COMPUTER SCIENCE AND INFORMATION TECHNOLOGY OSIJEK

Funded by
the European Union

Thank You!

Name: Sabina Žero

Organization:
UNSA

Email: sabinazero@pmf.unsa.ba

Website: www.unsa.ba; www.pmf.unsa.ba

NextAIRE

Funded by
the European Union